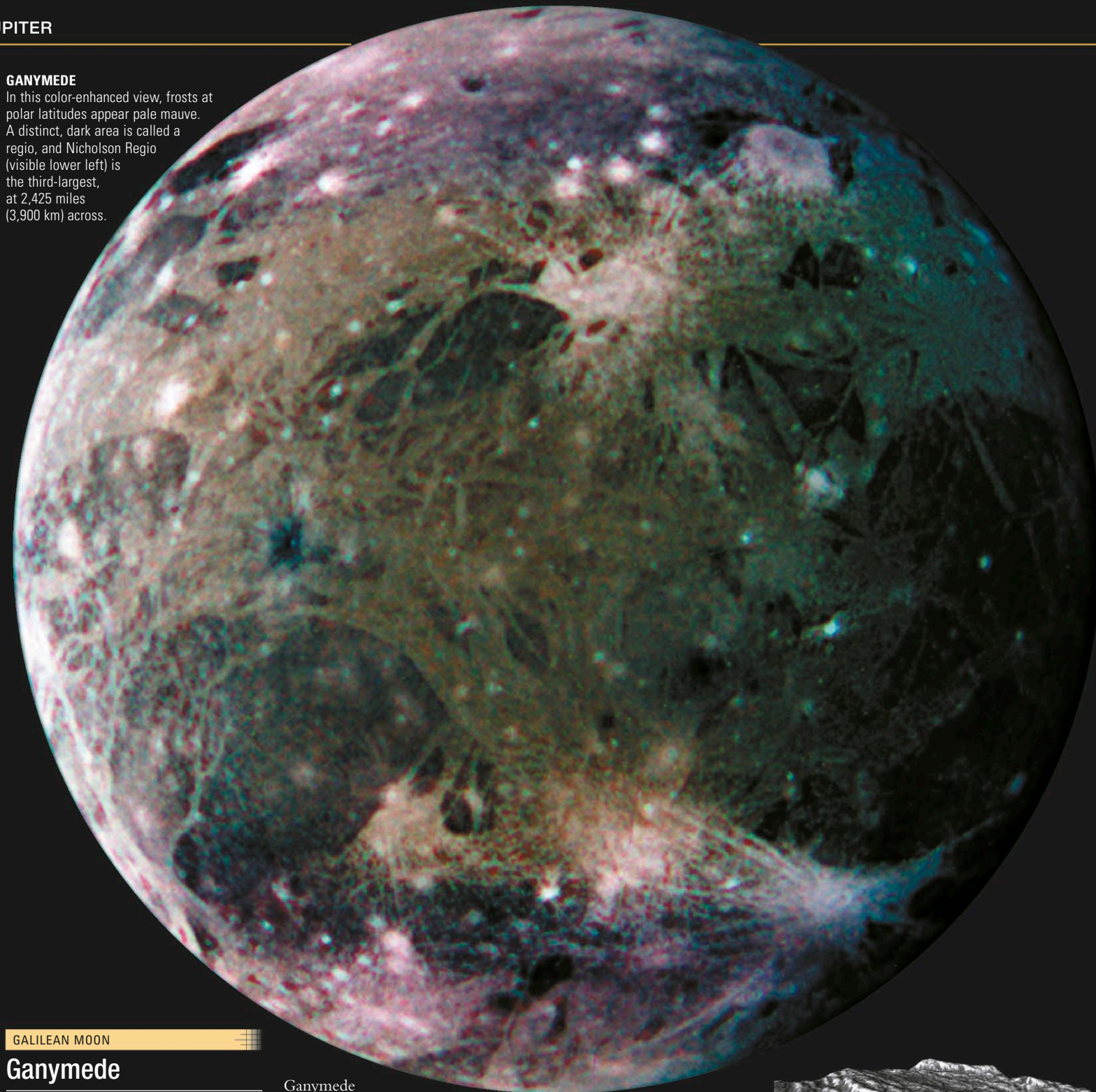


GANYMEDE

In this color-enhanced view, frosts at polar latitudes appear pale mauve. A distinct, dark area is called a regio, and Nicholson Regio (visible lower left) is the third-largest, at 2,425 miles (3,900 km) across.

**GALILEAN MOON****Ganymede**

DISTANCE FROM JUPITER 664,470 miles (1.07 million km)

ORBITAL PERIOD 7.15 Earth days

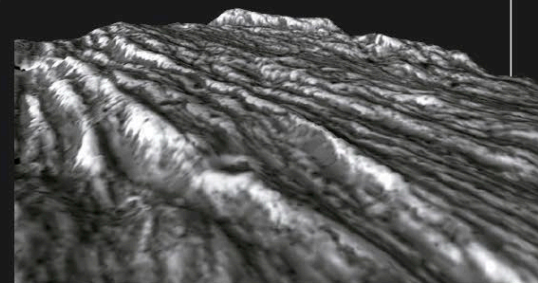
DIAMETER 3,267 miles (5,262 km)

Ganymede is the largest moon in the solar system, bigger than both Pluto and Mercury and three-quarters the size of Mars. It is named after the beautiful young boy in Greek myth who was taken to Olympus by Zeus and became cupbearer to the gods.

Ganymede was formed from a 60:40 mix of rock and ice. This differentiated, and today the moon has an iron-rich core surrounded by a lower mantle of rock, an upper mantle of ice, and an icy crust of contrasting dark and bright areas. The dark terrain is pockmarked by impact craters, suggesting that it is an older surface. Circular bright areas termed palimpsests are the smoothed-out and filled-in remains of craters formed on

URUK SULCUS

This computer-generated perspective shows the area Uruk Sulcus, named after a Babylonian city. Icy material can be seen on the crests of the parallel ridges. Sulcus is the term for the grooved and ridged regions of bright terrain.



the icy surface in the distant past. The dark terrain is also characterized by long depressions about 4 miles (7 km) wide, called furrows. These may have formed as subsurface ice flowed into recently formed craters and material dragged across the surface created the bow-shaped troughs.

The bright terrain, which is rich in water ice

with patches of carbon-dioxide ice, is generally smoother and marked by fewer craters. It is crisscrossed by ridges and grooves produced by the tectonic stretching of the moon's surface.

SIPPAR SULCUS

This depression within Sippa Sulcus appears to be an old caldera (a volcano's collapsed underground reservoir) containing frozen lava.

**INFRARED MAPPING**

The infrared image on the left, taken by Galileo, locates surface water ice—the brighter the shading, the greater the amount. The colors of the right-hand image indicate the location of minerals (red) and the size of ice grains (shades of blue).

